

# Linear mixed effects models

# Warmup

Work on the warmup activity (handout).

# Warmup

What are the groups in the data?

# Warmup

What mixed effects model should the researchers use to investigate their hypothesis?

# Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \stackrel{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

## + Shape:

- + the overall relationship between satisfaction and price is linear
- + The slope is the *same* for each neighborhood

# Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \stackrel{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

## + Independence:

- + random effects are independent
- + observations within neighborhoods are independent after accounting for the random effect (i.e., the random effect captures the correlation within neighborhoods)
- + observations from different neighborhoods are independent

# Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \stackrel{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

- + **Normality:** Both  $u_i \sim N(0, \sigma_u^2)$  and  $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$
- + **Constant variance:**
  - +  $\varepsilon_{ij}$  has the same variance  $\sigma_\varepsilon^2$  regardless of satisfaction or neighborhood

## Calculating predicted values

Residuals:  $Price_{ij} - \widehat{Price}_{ij}$ , where

$$\widehat{Price}_{ij} = \hat{\beta}_0 + \hat{\beta}_1 Satisfaction_{ij} + \hat{u}_i$$

Where do the  $\hat{u}_i$  come from?



## Estimated random effects

R calculates an estimated random effect for each group (i.e., neighborhood):

```
m1 <- lmer(price ~ overall_satisfaction +  
            (1 | neighborhood),  
            data = bnb)  
coef(m1)
```

```
## $neighborhood  
##               (Intercept) overall_satisfaction  
## Albany Park      16.367331      14.80912  
## Archer Heights   9.863461      14.80912  
## Avondale        40.533851      14.80912  
## Beverly         21.046464      14.80912  
## Bridgeport      13.304517      14.80912  
## Brighton Park   28.548742      14.80912  
## Burnside        12.741349      14.80912  
## Calumet Heights 11.465091      14.80912
```

# Estimated random effects

```
coef(m1)
```

|                | (Intercept) | overall_satisfaction |
|----------------|-------------|----------------------|
| Albany Park    | 16.367331   | 14.80912             |
| Archer Heights | 9.863461    | 14.80912             |
| Avondale       | 40.533851   | 14.80912             |
| Beverly        | 21.046464   | 14.80912             |
| Bridgeport     | 13.304517   | 14.80912             |

What is the same for every neighborhood?

# Estimated random effects

```
coef(m1)
```

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| Bridgeport     | 13.304517   | 14.80912             |

What is the same for every neighborhood?

The slope ( $\hat{\beta}_1$ )

## Estimated random effects

```
coef(m1)
```

|                | (Intercept) | overall_satisfaction |
|----------------|-------------|----------------------|
| Albany Park    | 16.367331   | 14.80912             |
| Archer Heights | 9.863461    | 14.80912             |
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What is *different* for each neighborhood?

# Estimated random effects

```
coef(m1)
```

|                | (Intercept) | overall_satisfaction |
|----------------|-------------|----------------------|
| Albany Park    | 16.367331   | 14.80912             |
| Archer Heights | 9.863461    | 14.80912             |
| Avondale       | 40.533851   | 14.80912             |
| Beverly        | 21.046464   | 14.80912             |
| Bridgeport     | 13.304517   | 14.80912             |

What is *different* for each neighborhood?

The intercept ( $\hat{\beta}_0 + \hat{u}_i$ )

## Estimated random effects

|                | (Intercept) | overall_satisfaction |
|----------------|-------------|----------------------|
| Albany Park    | 16.367331   | 14.80912             |
| Archer Heights | 9.863461    | 14.80912             |
| Avondale       | 40.533851   | 14.80912             |
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How do I get the random effect estimates  $\hat{u}_i$  ?

## Estimated random effects

|                | (Intercept) | overall_satisfaction |
|----------------|-------------|----------------------|
| Albany Park    | 16.367331   | 14.80912             |
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How do I get the random effect estimates  $\hat{u}_i$  ?

The intercept for each neighborhood is  $\hat{\beta}_0 + \hat{u}_i$

So subtract the estimate  $\hat{\beta}_0$  from the intercept for each neighborhood

# Checking assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \stackrel{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

**+ Normality assumption:** Both  $u_i \sim N(0, \sigma_u^2)$  and  $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$

How do you think we could check the normality assumption?



## QQ plots

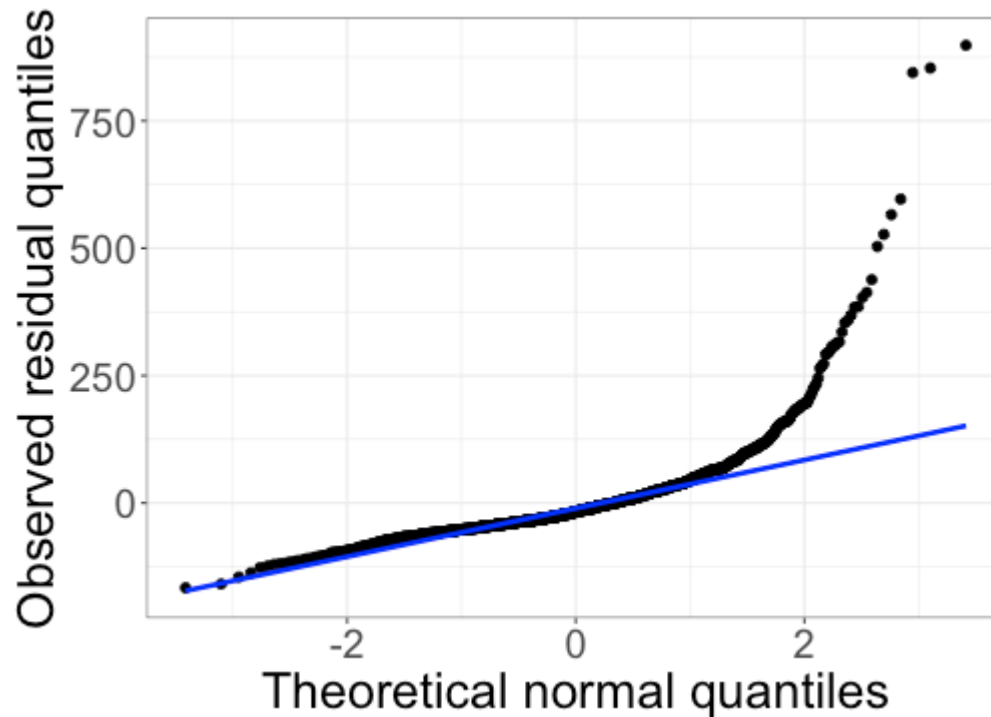
**Assumption:**  $u_i \sim N(0, \sigma_u^2)$

- + Check whether the random effect estimates  $\hat{u}_i$  appear normal with a QQ plot

**Assumption:**  $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$

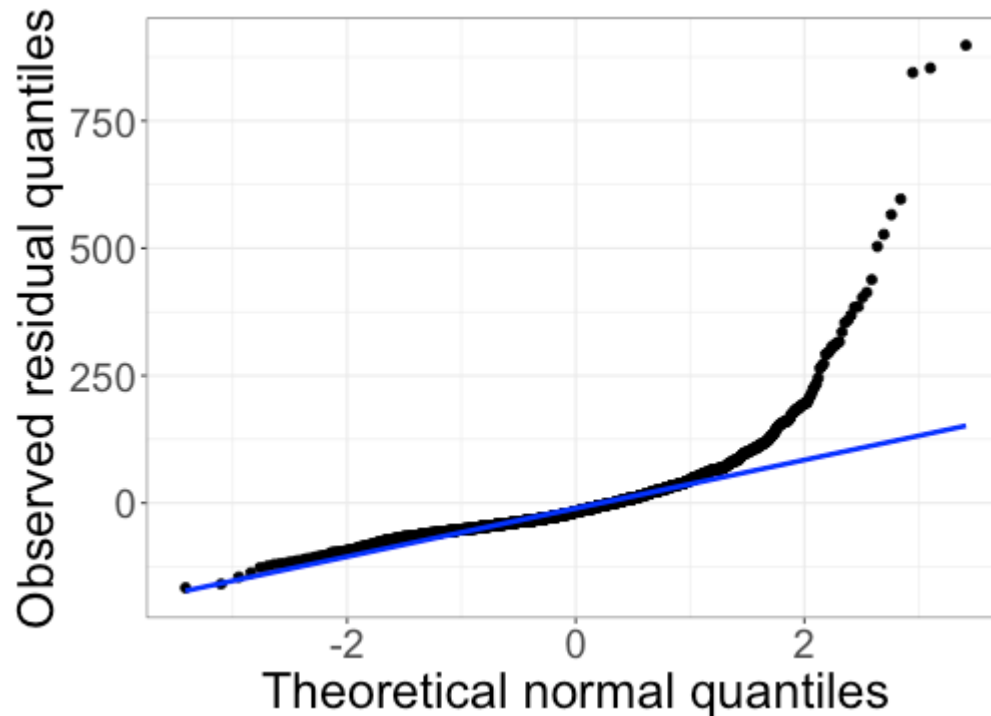
- + Check whether the residuals appear normal with a QQ plot

## QQ plot for the residuals



Do the residuals appear normal?

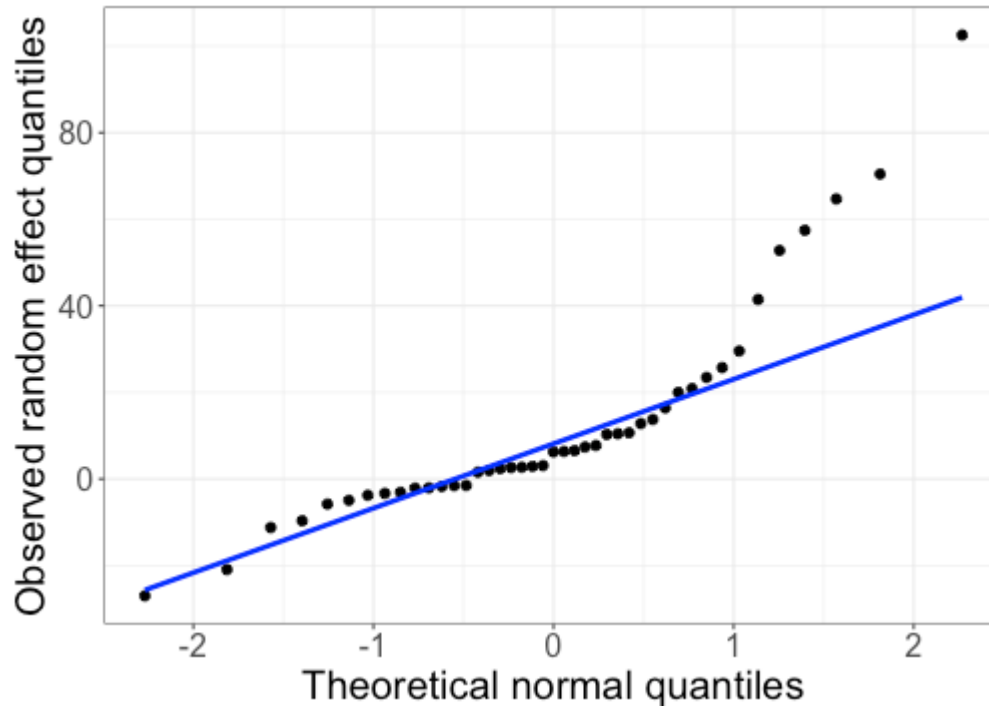
## QQ plot for the residuals



Do the residuals appear normal?

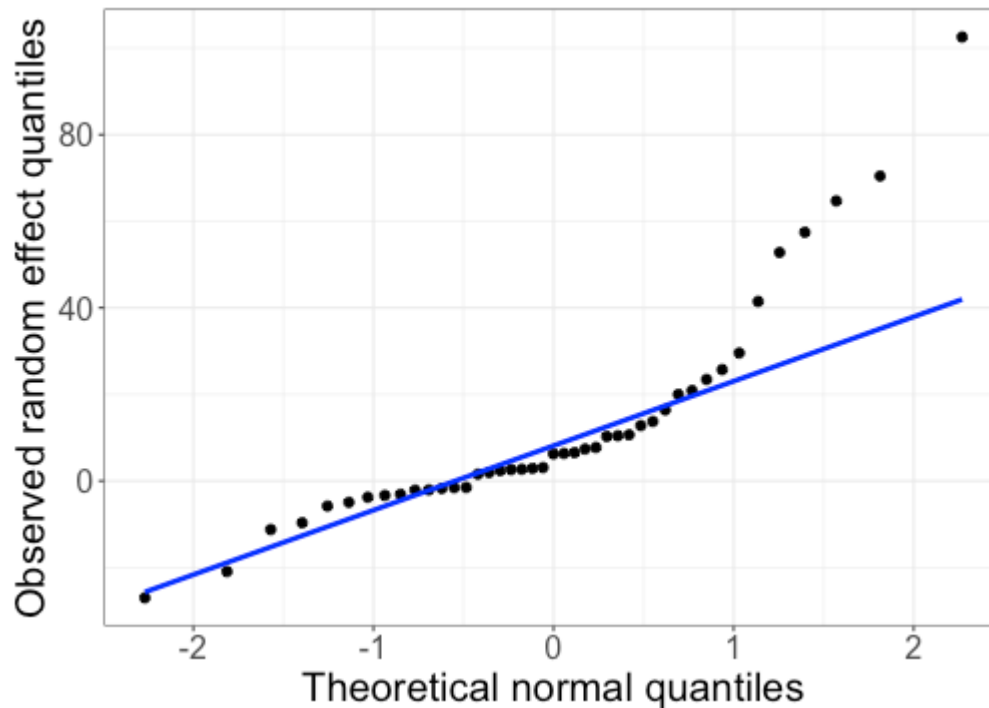
No -- though the sample size is large, so it might not matter

## QQ plot for the random effects



Do the random effects appear normal?

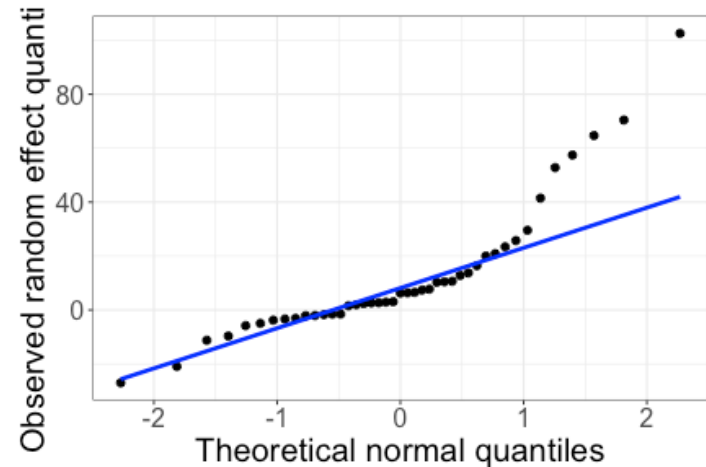
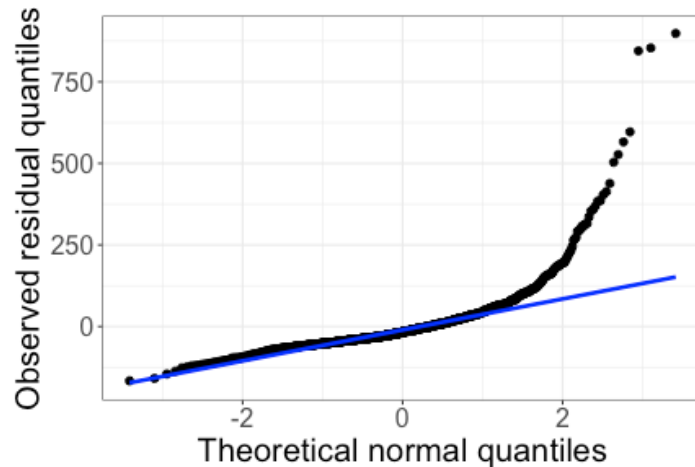
## QQ plot for the random effects



Do the random effects appear normal?

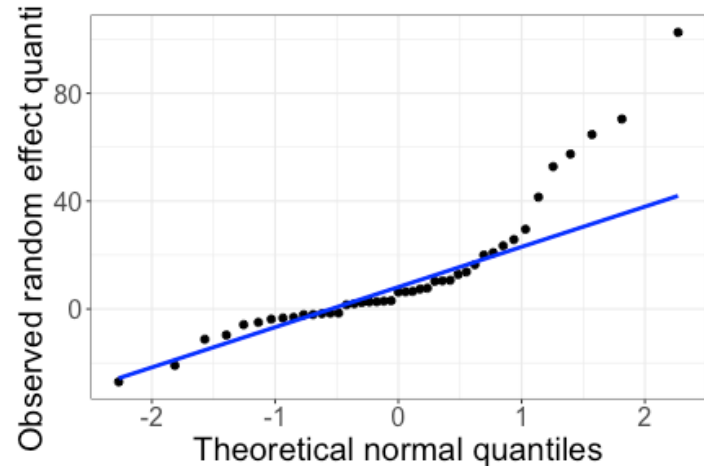
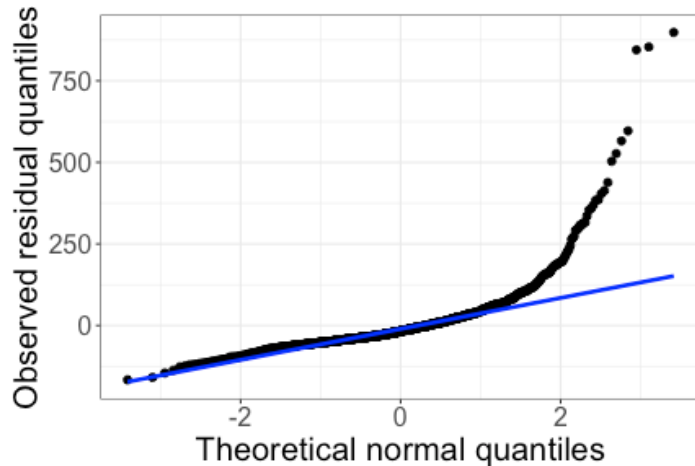
No -- and this could be more problematic

# Addressing assumption violations



How could we address violations of the normality assumptions?

# Addressing assumption violations



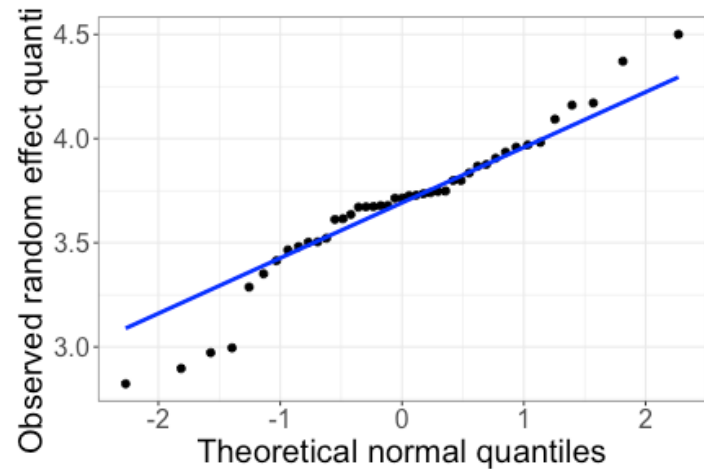
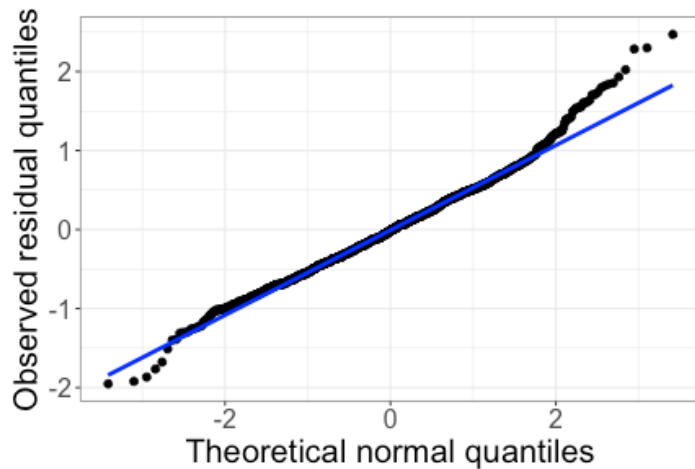
How could we address violations of the normality assumptions?

Transformations!

# Transformations

$$\log(\text{Price}_{ij}) = \beta_0 + \beta_1 \text{Satisfaction}_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \stackrel{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$



- ✚ Transformations can help with both the residuals and random effects



## Class activity

Work on the class activity on the course website. Submit your HTML on Canvas when finished.